

CONTACTS

Islamabad, PK
+92 311 5715112
swairarshad1@gmail.com

SKILLS

- AI & Voice: ElevenLabs Conversational AI, Canary Speech, Multi-agent Orchestration, LLM Integration
- AI-Assisted Development: Claude, ChatGPT, Cursor, Manus AI — daily use for code generation, debugging, architecture planning, and delivery acceleration
- Mobile: React Native, iOS Development, Android Development, Native Modules
- Languages: TypeScript, JavaScript
- State Management: Redux, Context API, Zustand
- UI & UX: Animations, Haptic Feedback, Figma collaboration
- Backend & APIs: RESTful APIs, Axios, Fetch, Firebase, AWS Amplify, MongoDB (familiarity)
- Payments & Auth: Apple In-App Purchases, Swish, Nets, Bank ID
- Specialisms: NFC, IoT, Offline Downloads, Push Notifications, App Store & Play Store deployment
- Forms & Validation: Formik
- Tools & Process: Git, Jira, Agile / Scrum, Project Management
- Ethical and Legal Compliances

EDUCATION

Nov 2022 – Feb 2023

PGD in Project Management, London College of Professional Studies (LCPS)

Sep 2015 – Jun 2019

BSc in Computer Science, PMAS Barani
Institute of Information Technology |
Rawalpindi, Pakistan

LANGUAGES

Urdu — Native / Bilingual
English — Full Professional Proficiency
French — Elementary

LINKS

[Portfolio](#)
[LinkedIn](#)
[GitHub](#)
[Medium](#)
[Dev.to](#)

Swair Arshad

Senior Full-Stack Mobile Engineer & AI Integration Specialist

PROFESSIONAL SUMMARY

Senior Full-Stack Mobile Engineer & AI Integration Specialist at DPL with 6+ years delivering production-grade iOS and Android applications end-to-end, from architecture and native module development (Swift, Java/Kotlin) through performance optimization, App Store and Play Store release, to post-launch iteration. Led small agile teams, and delivered across 9+ client projects, mentoring engineers, leading code reviews, and driving architectural decisions while collaborating cross-functionally with design, product, QA, and backend teams. Delivered a 30% average performance improvement and 95% on-time delivery rate across all engagements.

Specializes in integrating AI capabilities into mobile, including real-time conversational AI, LLM-powered pipelines, and vocal biomarker analysis. Also contributed in backend capacity on feature development and maintenance. Leverages AI tooling daily (Claude, Cursor, ChatGPT) to accelerate development velocity and code quality.

PROFESSIONAL EXPERIENCE

Senior Full-Stack Mobile Engineer — DPL

Dec 2019 – Present

Islamabad, Pakistan

- Led 2–3 person agile teams across 9+ client projects — sprint planning, code reviews, and client-facing communication from requirements to App Store release.
- Delivered a 30% improvement in app performance and a 95% on-time delivery rate across all engagements.
- Built AI-powered features for NOVA LiFE: real-time sobriety companion with ElevenLabs Conversational AI voice pipelines, vocal biomarker analysis via Canary Speech SDK, and multi-agent orchestration — achieving sub-500ms end-to-end latency.
- Integrated payment gateways (Swish, Nets, Apple IAP), authentication (Bank ID), NFC, IoT, Firebase, and AWS Amplify across multiple production apps.
- Mentored junior developers on React Native best practices, state management patterns, and mobile performance optimisation.
- Featured by DPL in their company success story series.

KEY PROJECTS

MunshiJi — AI Personal Finance App

Jan 2026 — Present

React Native (Expo) · TypeScript · Node.js/Express · Drizzle ORM · PostgreSQL · Firebase Auth · Google Gemini

- Building an AI-first personal finance app end-to-end across mobile, backend, and admin portal — voice-driven expense logging (speech-to-structured-expense via LLM), budgeting, and an in-app AI chat assistant.
- Architected the Node.js/Express/TypeScript backend (Drizzle ORM, PostgreSQL, JWT + Firebase auth, rate limiting, Swagger-documented REST API) alongside the React Native mobile client and a Vite/React admin portal.
- Led the app toward a cohort-1 test launch, driving scope and architecture calls under a hard feature freeze — including graceful offline handling, and a phased rollout of investment features ahead of regulatory review.
- Enforced a strict TypeScript, enum-driven, i18n-first codebase across mobile, backend, and admin surfaces.

ScanBazar — AI Halal/Haram Product Scanner

Jul 2026 — Present

React Native · Node.js/Express · PostgreSQL · Google Gemini

- Architecting a multi-stage AI pipeline that scans food/product labels and returns halal/haram/makruh verdicts: barcode and perceptual-hash matching first, falling through to a Gemini multimodal pass that extracts a clean ingredient list plus origin/manufacture data in one call, filtering marketing and nutrition text without a separate rules-based step.

- Engineering the ingredient-matching and lookup layer: normalizing and translating extracted text, parsing it into discrete ingredients/E-numbers, and matching against a curated database, with origin/country data cross-referenced to help resolve doubtful (mashbooh) cases.
- Designing classification logic so database-sourced verdicts return verbatim, unmatched/contested ingredients route to a scholar-review queue, and barcode/image-hash caching makes repeat scans instant.
- Layering in certification-logo detection as a secondary fast-path signal.

NOVA LiFE — AI Sobriety Companion*Jan 2026 — Jun 2026*

React Native · ElevenLabs Conversational AI · Canary Speech · Zustand

- Built an AI-powered mental wellness app delivering real-time therapeutic voice conversations through a custom ElevenLabs Conversational AI integration, with seamless switching between voice and text modalities.
- Engineered a full system managing session lifecycle from connect through multi-agent workflow transfers to graceful teardown.
- Embedded Canary Speech SDK for real-time vocal biomarker analysis (mood, stress, energy), surfacing acoustic health signals to the AI as live contextual updates mid-conversation.
- Built mobile-side orchestration for ElevenLabs multi-agent workflows including silent agent transfers, dynamic variable injection, and partial state recovery on mid-session exit.

The Breath Source

April 2025 — Dec 2025

React Native · Firebase · AWS Amplify

- Built a subscription-based wellness app with curated breathwork sessions, offline downloads, live class streaming, personalised flows, and an integrated bio-hacking shop with in-app purchases. Live on App Store & Play Store.

DQCA — Quranic Script Certification

Jan 2024 — Dec 2024

- Developed a mobile platform to authenticate and certify Quranic scripts through verified huffaz, automating a previously manual certification process with robust online/offline support.

0-100 — Quiz Game

React Native · Firebase · Redux · Formik · Animatable

- Fast-paced quiz game. Integrated Apple In-App Purchases and optimised UI animations. Live on App Store.
- Led end-to-end development and optimization of consumer trivia game features, driving a 30% increase in user engagement through data-driven design and iterative testing.

Healness & Co · nGAGE @ Work · Stegvis · Switchpoint · GSTAR2020

React Native · Swish · Nets · Bank ID · NFC · IoT · Scrum

- Range of production apps: on-demand massage booking with Swish/Bank ID payments, enterprise employee feedback platform, clinical eating disorder diary, NFC tag communication tool, and an international conference app — all shipped to App Store and/or Play Store.

ACHIEVEMENTS & COMMUNITY

Rebel Stories — DPL Success Story Feature

Featured by DPL on their company channel and socials as part of their Rebel Stories series, spotlighting standout team members.

Toastmasters International — Club President (3 Terms)

Served three consecutive terms as President of Rebel Speakers, leading members in public speaking, leadership development, and communication skills. Also served as Master of Ceremonies at a Toastmasters District event.

University Guest Speaker — Tech & Career Inspiration

Invited by a university to deliver a session for new joiners on technology careers, inspiring students entering the software industry.

Government Coding Bootcamp — Lead Instructor

Designed and taught a 10-day intensive programming course for summer students at a government school through DPL, helping young learners discover their interest in coding and software development.

EmpowerHer — Women in Tech Speaker

Delivered and led sessions on leadership and technology as part of the EmpowerHer initiative, advocating for greater representation of women in the tech industry.

Technical Blogger

Active author on Medium and Dev.to, writing about mobile engineering and AI integration.